

‘A Fine Balance’: Exploring the Interplay of Reproductive Strategies and Growth Dynamics in Trees

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Masting is the phenomenon of synchronized and highly variable seed production within plant populations. It plays a critical role in ecosystem-level functions, influencing both plant regeneration and the population dynamics of seed predators. Despite its importance, models of masting are poorly predictive, likely because of the complex temporal dependencies and the difficulty of integrating environmental variation with internal plant dynamics. One of the most important internal dynamics hypothesized to drive masting is a trade-off between vegetative growth (leaves, stems, wood) and reproduction (seeds). However, this trade-off is often obscured by environmental variation simultaneously. Here, we leverage a rarely available coupled dataset over time and space, along with a new hierarchical Bayesian generative model, to test for evidence of this trade-off while accounting for variation in the environment.

The model explicitly treats annual resource allocation as a latent process occurring at the individual tree level, informed by year-to-year environmental variation and two distinct data sources: annual tree-ring widths as a proxy for growth and annual seed production records as a proxy for reproduction, collected from Mount Rainier National Park since 2008. Our approach reconstructs hidden resource pools from observed ecological data and tests how they shift in response to environmental variation during masting events, specifically modeling shifts in snowpack that shape moisture over the growing season and temperatures that drive growing season warmth. Uniquely, our dataset includes multiple stands across an elevational gradient, allowing us to model how environmental variation both between years and across space within years may shape resource pools and allocation. We utilize hierarchical structures to pool information across individuals, stands, and species to help estimate latent state transitions and how they vary across scales. This model demonstrates the use of Bayesian modeling and latent variable frameworks helping decipher unobservable physiological processes from observable ecological data.